

Last time:

CRT formulated in the version $R \rightarrow R/I \times R/J$
where I and J are coprime ideals in R (coprime means $I + J = R$).

$$\varphi : R \rightarrow R/I \times R/J$$

$$\varphi(r) = ([r]_I, [r]_J)$$

is a surjective homomorphism

$\Leftrightarrow$ means the system of congruences

$$\begin{cases} r \equiv y \pmod{I} \\ r \equiv z \pmod{J} \end{cases}$$

in the unknown $r \in R$ is solvable for any given right hand sides $y, z \in R$.

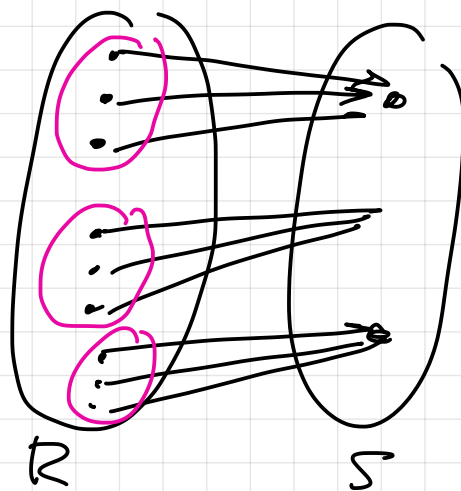
In the proof you need a representation

$$1 = \underbrace{u}_{\in I} + \underbrace{v}_{\in J}$$

1.7.8 Every ring homomorphism $\varphi: R \rightarrow S$
can be converted to an injective ring homomorphism

$$\psi : R/\ker(\varphi) \rightarrow S \text{ with}$$

$$\psi([r]_{\ker(\varphi)}) = \varphi(r)$$



1.7.9 Theorem (CRT - version $R/I \cap J \rightarrow R/I \times R/J$)

Let I and J be coprime ideals in a ring R . Then the map $\psi: R/I \cap J \rightarrow R/I \times R/J$, given

$$\text{by } \psi([r]_{I \cap J}) := ([r]_I, [r]_J),$$

is an isomorphism. In particular,

$$R/I \cap J \cong R/I \times R/J.$$

Proof: First, note that ψ is well defined, because $I \cap J \subseteq I$ and $I \cap J \subseteq J$. So it is well-defined homomorphism. By the previous version of the CRT (1.7.7), we know that ψ is surjective. To verify that ψ is injective it is enough to check that $\ker \psi = \{0\}$.

$$\psi([r]_{I \cap J}) = 0_{R/I \times R/J} \iff$$

$$[r]_I = 0_{R/I} \text{ and } [r]_J = 0_{R/J} \iff$$

$$r \equiv 0 \pmod{I} \text{ and } r \equiv 0 \pmod{J} \iff$$

$$r \in I \text{ and } r \in J \iff$$

$$r \in I \cap J \iff$$

$$r \equiv 0 \pmod{I \cap J} \iff$$

$$[r]_{I \cap J} = 0_{R/(I \cap J)}.$$

□

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1.7.10 Example $R = \mathbb{Z}$, $I = 4\mathbb{Z}$, $J = 7\mathbb{Z}$

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$$I + J = \langle 4, 7 \rangle = \langle \gcd(4, 7) \rangle = \langle 1 \rangle = \mathbb{Z}$$

$\Rightarrow I$ and J are coprime.

$$\gamma: \mathbb{Z}/I \cap J \rightarrow \mathbb{Z}/I \times \mathbb{Z}/J$$

$$\gamma([r]_{I \cap J}) = ([r]_I, [r]_J) \text{ is an isomorphism.}$$

$$I \cap J = 4\mathbb{Z} \cap 7\mathbb{Z} = 28\mathbb{Z}. \text{ Explicitly:}$$

$$\gamma: \mathbb{Z}/28 \rightarrow \mathbb{Z}/4 \times \mathbb{Z}/7$$

$$\gamma([r]_{28}) = ([r]_4, [r]_7).$$

Let's invert this isomorphism.

Effectively, we solve a system of congruences

$$\begin{cases} r \equiv a \pmod{4} \\ r \equiv b \pmod{7} \end{cases}$$

$a, b \in \mathbb{Z}$ given and r is unknown.

$$\begin{aligned} \exists! A \leadsto 1 &= \gcd(4, 7) = 2 \cdot 4 + (-1) \cdot 7 \\ &= \underbrace{8}_{\equiv 1} + \underbrace{(-7)}_{\equiv 0}. \end{aligned}$$

$$\begin{aligned} -7 &\equiv 1 \pmod{4} \\ 8 &\equiv 0 \pmod{4} \end{aligned}$$

$$\begin{aligned} -7 &\equiv 0 \pmod{7} \\ 8 &\equiv 1 \pmod{7} \end{aligned}$$

$$r \equiv a \cdot (-7) + b \cdot 8 \pmod{28}$$

This describes the complete solution set of our system of congruences.

The inverse isomorphism is

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$$\psi^{-1}([a]_4, [b]_2) = [a \cdot (-7) + b \cdot 8]_{28}$$

In the next version of the CRT, we have $m \geq 2$ ideals rather than just two.

1.7.11 A Lemma on coprimality. If in a ring R ,

an ideal I_1 is coprime to ideals $I_2, \dots, I_m$

where $m \in \mathbb{Z}_{\geq 2}$ (which means $I_1 + I_j = R$

for every $j = 2, \dots, m$), then I_1 is also coprime

to $I_2 \cap \dots \cap I_m$.

Proof: Since $R = I_1 + I_j$ for each $j = 2, \dots, m$,

we have $1 = u_j + v_j$ for some $u_j \in I_1$ and $v_j \in I_j$.

We take the product:

$$1 = \prod_{j=2}^m (u_j + v_j)$$

Now, we can expand the brackets on the right hand side. Say for $m = 3$:

$$1 = (u_2 + v_2)(u_3 + v_3)$$

$$= \underbrace{u_2 u_3 + u_2 v_3 + v_2 u_3}_{\in I_1} + \underbrace{v_2 v_3}_{\substack{\cap \\ I_2 \cap I_3}}$$

When we expand brackets in general, we have 2^{m-1} terms, where each term contains

some u_j as a factor belonging to I_1 . (5)

The only term that contains none of the $u_2, \dots, u_m$ as a factor is the term

$$v_2 \cdot \dots \cdot v_m. \text{ Since } u_j \in I_j \text{ for every}$$

$j=2, \dots, m$ the term $u_2 \cdot \dots \cdot u_m$ belongs to

each of the ideals $I_2, \dots, I_m$ and by this to

their intersection $I_2 \cap \dots \cap I_m$. We conclude

that $1 = a_1 + b_1$ where $a_1 \in I_1$

and $b_1 = v_2 \cdot \dots \cdot v_m \in I_2 \cap \dots \cap I_m$.

So, $I_1 + (I_2 \cap \dots \cap I_m)$ contains 1.

$$\Rightarrow I_1 + (I_2 \cap \dots \cap I_m) = R. \quad \square$$

1.7.12 The previous assertion can be strengthened to I_1 and $I_2 \cdot \dots \cdot I_m$ being coprime.

1.7.13 Theorem

(CRT - version $R \rightarrow R/I_1 \times \dots \times R/I_m$)

Let $I_1, \dots, I_m$ be pairwise coprime ideals in a ring R , where $m \in \mathbb{Z} \geq 2$. That means,

$I_i + I_j = R$ for $i \neq j$. Then the map

$$\varphi: R \rightarrow R/I_1 \times \dots \times R/I_m,$$

$$\varphi(r) = ([r]_{I_1}, \dots, [r]_{I_m}),$$

is a surjective homomorphism.

Proof: Clearly, φ is a homomorphism of rings. We need to show that it is surjective. That means, we need to show that the system of congruences

$$\begin{cases} r \equiv y_1 \pmod{I_1} \\ \vdots \\ r \equiv y_m \pmod{I_m} \end{cases}$$

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in the unknown $r \in R$ has a solution, no matter how we choose $y_1, \dots, y_m \in R$.

By Lemma 1.7.11, the ideal I_i is coprime with the ideal $\bigcap_{j \in \{1, \dots, m\} \setminus \{i\}} I_j$. So, we can represent

$1 \in R$ as $1 = a_i + b_i$ with

$a_i \in I_i$ and $b_i \in \bigcap_{j \in \{1, \dots, m\} \setminus \{i\}} I_j$ (that is,

$b_i \in I_j$ when $j \neq i$). So,

$b_i = 1 - a_i \equiv 1 \pmod{I_i}$ and

$b_i \equiv 0 \pmod{I_j}$ when $j \neq i$.

$\Rightarrow r = y_1 \cdot b_1 + y_2 \cdot b_2 + \dots + y_m \cdot b_m$

solves the system of the congruences. $\square$

1.7.14 Theorem (CRT - Version)
 $R / I_1 \cap \dots \cap I_m \longrightarrow R / I_1 \times \dots \times R / I_m$.

Let $I_1, \dots, I_m$ be pairwise coprime ideals in a ring R , where $m \in \mathbb{Z}_{\geq 2}$. Then, the map

$\psi : R / I_1 \cap \dots \cap I_m \longrightarrow R / I_1 \times \dots \times R / I_m$,

$$\psi([r]_{I_1 \cap \dots \cap I_m}) = ([r]_{I_1}, \dots, [r]_{I_m}) \quad (7)$$

is a ring isomorphism. In particular,

$$R / I_1 \cap \dots \cap I_m \cong R / I_1 \times \dots \times R / I_m.$$

Proof: Direct consequence of 1.7.13.

Checking that $\ker(\psi) = \{0\}$ to show injectivity is straightforward. $\square$

1.7.15 Example. $R = \mathbb{Z}$, $I_1 = 2\mathbb{Z}$, $I_2 = 3\mathbb{Z}$, $I_3 = 5\mathbb{Z}$

$$\left. \begin{aligned} I_1 + I_2 &= 2\mathbb{Z} + 3\mathbb{Z} = \mathbb{Z} \\ I_1 + I_3 &= 2\mathbb{Z} + 5\mathbb{Z} = \mathbb{Z} \\ I_2 + I_3 &= 3\mathbb{Z} + 5\mathbb{Z} = \mathbb{Z} \end{aligned} \right\} \text{assumptions of the CRT are satisfied}$$

$$\psi : \mathbb{Z} / I_1 \cap I_2 \cap I_3 \rightarrow \mathbb{Z} / I_1 \times \mathbb{Z} / I_2 \times \mathbb{Z} / I_3$$

$$\psi([r]_{I_1 \cap I_2 \cap I_3}) = ([r]_{I_1}, [r]_{I_2}, [r]_{I_3})$$

is a ring isomorphism (by 1.7.14).

$$I_1 \cap I_2 \cap I_3 = 2\mathbb{Z} \cap 3\mathbb{Z} \cap 5\mathbb{Z} = 30\mathbb{Z}.$$

$$\psi : \mathbb{Z} / 30 \rightarrow \mathbb{Z} / 2 \times \mathbb{Z} / 3 \times \mathbb{Z} / 5$$

$$\psi([r]_{30}) = ([r]_2, [r]_3, [r]_5)$$

We are solving a system of congruences

$$\begin{cases} r \equiv a \pmod{2} \\ r \equiv b \pmod{3} \\ r \equiv c \pmod{5} \end{cases}$$

$a, b, c \in \mathbb{Z}$ given, r is unknown.

We know $2\mathbb{Z}$ and $3\mathbb{Z} \cap 5\mathbb{Z} = 15\mathbb{Z}$ are coprime
 $3\mathbb{Z}$ and $2\mathbb{Z} \cap 5\mathbb{Z} = 10\mathbb{Z}$ are coprime, too
 $5\mathbb{Z}$ and $2\mathbb{Z} \cap 3\mathbb{Z} = 6\mathbb{Z}$ are also coprime.

We can use FEA to produce the following:

$$1 = -14 + 15 \text{ with } -14 \in 2\mathbb{Z} \text{ and } 15 \in 3\mathbb{Z} \cap 5\mathbb{Z}$$

$$1 = -9 + 10 \text{ with } -9 \in 3\mathbb{Z} \text{ and } 10 \in 2\mathbb{Z} \cap 5\mathbb{Z}$$

$$1 = -5 + 6 \text{ with } -5 \in 5\mathbb{Z} \text{ and } 6 \in 2\mathbb{Z} \cap 3\mathbb{Z}$$

The solution of the system of the congruences is

$$x = 15 \cdot a + 10 \cdot b + 6 \cdot c$$

The inverse of ψ or ψ^{-1}

$$\psi^{-1}([a]_2, [b]_3, [c]_5) = [15a + 10b + 6c]_{30}$$

